

YT-Agent Project Code Review (v2)

Generated: June 19, 2026 | Agents 1-4 built

Issues Found

#	File	Line	Issue	Sev.
1	ui/app.py	all results	Writer drafts, research briefs, and failed_formats are never	High
2	ui/app.py	sidebar ~22	Displays Brief Agents 3 & 4 as not built (shows x2 empty) but	Medium
3	agents/research_agent.py	~107	def _parse_brief(...) -> dict None: uses Python 3.10+ union	Medium
4	agents/writer_agent.py	~115	def _write_brief(...) -> str None: uses Python 3.10+ union	Medium
5	agents/research_agent.py	~63	failed_briefs dict is created and populated in the loop but never	Low
6	agents/writer_agent.py	~145	State in str(e).lower() uses DeadVariable: inconsistent with the	Low
7	agents/research_agent.py	~120	Same broad rate-limit check: "rate" in str(e).lower() - should be	Low
8	agents/orchestrator.py	53	Catchment.says "Stages 3-6 not built yet (placeholders)" - now	Low
			wrong. Stages 3 and 4 are live.	

Cross-Check Results

Check	Result
ui/app.py imports from agents/orchestrator.py	PASS
orchestrator.py imports ResearchAgent and WriterAgent	PASS
orchestrator._run_writer_stage() accesses writer_result.failed_formats	PASS
WriterResult.failed_formats exists in base.py	PASS
All prompt files have {brief} placeholder	PASS
PROMPTS_DIR in writer_agent.py correctly points to prompts/	PASS
requirements.txt has streamlit, groq, yt-dlp, python-dotenv, pydub	PASS
railway.toml has PYTHONPATH=/app in start command	PASS
nixpacks.toml installs ffmpeg	PASS
No hardcoded Windows paths that break on Railway Linux	PASS
No relative imports that break on Railway	PASS
research_agent.py uses dict None (Python 3.10+ only)	WARN
writer_agent.py uses str None (Python 3.10+ only)	WARN
ui/app.py shows writer drafts to the user	FAIL
ui/app.py sidebar reflects Agents 3 & 4 as built	FAIL
reviewer_agent.py and publisher_agent.py exist	N/A - safe

Priority Fix List

Fix 1 - High | ui/app.py (line results section)

Add display of research briefs, writer drafts, and failed_formats after the planner section.

Add after planner section:

```
if result.writer_result and result.writer_result.drafts:
    for fmt, draft in result.writer_result.drafts.items():
        with st.expander(f"{fmt}"):
            st.markdown(draft)
```

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Fix 2 - Medium | ui/app.py (line sidebar ~22)

Update sidebar checkboxes to show Agents 3 & 4 as built.

```
- checkmark **Agent 3** - Research
- checkmark **Agent 4** - Writer
```

Fix 3 - Medium | agents/research_agent.py (line ~107)

Remove dict | None return annotation for Python 3.9 compatibility.

```
def _parse_brief(self, raw: str):
```

Fix 4 - Medium | agents/writer_agent.py (line ~115)

Remove str | None return annotation for Python 3.9 compatibility.

```
def _write_format(self, fmt, transcript, title, brief, config):
```

Fix 5 - Low | agents/research_agent.py (line ~63)

Remove dead failed_briefs variable (never stored or returned).

```
Delete: failed_briefs = {} and all failed_briefs[fmt] = ... assignments
```

Fix 6 - Low | agents/writer_agent.py (line ~145)

Tighten inline rate-limit check to match error_classifier fix.

```
if "429" in str(e) or "rate limit" in str(e).lower():
```

Fix 7 - Low | agents/research_agent.py (line ~120)

Same inline rate-limit tightening.

```
if "429" in str(e) or "rate limit" in str(e).lower():
```

Fix 8 - Low | agents/orchestrator.py (line 53)

Update stale comment.

```
# Stages 3-4 live; Stages 5-6 not built yet
```

Current Build Status

Stage	Class	File	Status
Agent 1	TranscriptAgent	agents/transcript_agent.py	BUILT
Agent 2	PlannerAgent	agents/planner_agent.py	BUILT
Agent 3	ResearchAgent	agents/research_agent.py	BUILT
Agent 4	WriterAgent	agents/writer_agent.py	BUILT
Agent 5	ReviewerAgent	agents/reviewer_agent.py	NOT YET
Agent 6	PublisherAgent	agents/publisher_agent.py	NOT YET